EPSR-test-ex5 Complete reset (ireset 1)

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the complete reset of an EPSR [1,2] simulation, `ireset 1` in the `.EPSR.inp` file: the empirical potential, the automatic `ereq` control and the accumulated averages are discarded and default pair potentials are generated, which users do to restart a refinement from the reference potential only [3].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (`pressreq/ereqstep/erequnit` := 0, reset of `ireset` after the run), `inita` (accumulators restarted, `.inpa` not read)
- `read_and_save_files.f`: `read_pcof` (EP coefficients reset, default pair potentials)
- `molfit.f`: `molfit` (no step-size adaptation while `ireset` = 1)

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::setup`, `Epsr::iterate`
- `structure.rs`: `State::inita`, `reset_accumulators`
- `pcof.rs`: `read_pcof`, `default_pair_potentials`; `montecarlo.rs`: `molfit`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. After the reset the run continues with zero EP amplitude, i.e. the configuration evolves under the reference potential alone. The pair table is rebuilt from the atom types of the `.ato` (21 pairs) with zero EP coefficients and, as `rminfac` is 0, no repulsive cores.

Inputs

Copy of EPSR-test with only `ireset 1` (instead of 0) in `200DMSO400Water.EPSR.inp`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Three chained passes. Pass 1 performs the reset and the Fortran writes `ireset 0` back to the `.inp`, so passes 2-3 test the normal continuation from reset files. Evidence in the stored Fortran reference: `ereq` = 0 and `ereqstep 0.000 0.000` in the final `.pcof`, the `.erg` history restarted (3 lines instead of 940 + 3), and no `delfit.dat` written (no EP fit). The 26 files written are identical after every pass.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.